

Kasra Ferdowsi

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Engineering Experience

Staff Software Engineer <i>Cubist Inc</i>	2025 - Present
Senior Software Engineer <i>Cubist Inc</i>	2024 - 2025

- Contributed over 250 pull requests spanning Rust back-end services, AWS CDK infrastructure, TypeScript and Rust SDKs, WebAssembly policies, example programs, and documentation.
- Designed and built an Access Control (ACL) system in Rust providing fine-grained user-friendly control over sensitive resources such as CubeSigner keys, policies, and secrets.
- Productionized and maintained the Cubist Policy SDK; Built its CI/CD pipeline, fixed bugs and outdated dependencies, and extended it with JSON and gRPC clients and other features.
- Developed the CubeSigner RPC API for transaction management and submission, including fully-concurrent EVM nonce management.
- Led design, requirements gathering, and implementation for the second iteration of the *Bascule Drawbridge*; Securing the transfer of millions of dollars in funds from malicious actors and bugs.

Graduate Student Researcher <i>UC San Diego</i>	2019 - 2024
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- Invented novel techniques and interaction models for using bottom-up enumerative techniques for efficient synthesis of Python programs.
- Maintained the Projection Boxes live programming environment, extended it with program synthesis features, and ported it to a web editor with a Java backend.
- Ran multiple user studies with up to hundreds of participants to evaluate the usability of programming tools, and visualized and performed statistical analysis on the resulting study data.

Research Intern <i>Microsoft Research</i>	2022
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- Designed and implemented ColDeco, a novel interaction model for inspecting and diagnosing errors in AI-generated code by end-users.

Software Engineer Intern <i>Tom Sawyer Software</i>	2016 - 2019
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- Helped design and develop the Tom Sawyer Licensing Server 2.0, a backwards-compatible scalable web server, in Java using a micro-service architecture.
- Implemented account management, and a web interface for Graph Database Browser.
- Co-designed and developed a tool combining Git, Maven and Docker to streamline building and deploying Tom Sawyer Perspectives applications.

Education

UC San Diego, California, USA	2024
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- PhD in Computer Science. Advised by Nadia Polikarpova and Sorin Lerner.
- M.S. in Computer Science.

UC Berkeley, California, USA	2019
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- B.S. in Electrical Engineering and Computer Science.
- Graduated with Highest Honors.

Expertise

Programming Languages

- Advanced knowledge of Rust and Python. Fluent in TypeScript, Scala, and Java. Familiar with Haskell, JavaScript, Kotlin, and Go.

DevOps & Infrastructure

- Advanced knowledge of DynamoDB. Fluent in Git and GitHub Actions. Experience working with Docker, AWS CDK, EC2, Lambdas, Secrets Manager, and other AWS services.

Publications

Bottom-up Synthesis of Memory Mutations with Separation Logic <i>Ferdowsi, K., Peleg, H.</i>	ECOOP 2025
Human-Centered Program Synthesis <i>Ferdowsifard, K.</i>	Dissertation
Validating AI-Generated Code with Live Programming <i>Ferdowsi, K., Huang, R., James, M. B., Polikarpova, N., and Lerner, S.</i>	CHI 2024
ColDeco: An End User Spreadsheet Inspection Tool for AI-Generated Code <i>Ferdowsi, K., Williams, J., Drosos, I., Gordon, A.D., Negreanu, C., Polikarpova, N., Sarkar, A., and Zorn, B.</i>	VL/HCC 2023
Towards Human-Centered Types and Type Debugging <i>Ferdowsi, K.</i>	PLATEAU 2023
The Usability of Advanced Type Systems: Rust as a Case Study <i>Ferdowsi, K.</i>	Research Exam
Investigating the Impact of Using a Live Programming Environment in a CS1 Course <i>Huang, R., Ferdowsi, K., Selvaraj, A., Soosai Raj, A.G., and Lerner, S.</i>	SIGCSE TS 2022
LooPy: Interactive Program Synthesis with Control Structures <i>Ferdowsifard, K., Barke, S., Peleg, H., Lerner, S., and Polikarpova, N.</i>	OOPSLA 2021
Small-Step Live Programming by Example <i>Ferdowsifard, K., Ordookhanians, A., Peleg, H., Lerner, S., and Polikarpova, N.</i>	UIST 2020

Teaching Experience

Teaching Assistant <i>CSE130: Programming Languages</i> - Prepared and presented discussion sections, held office hours, and graded assignments. - Also taught in Winter 2022 and Fall 2021.	Fall 2023
Teaching Assistant <i>CSE230: Graduate Programming Languages</i> - Prepared and presented discussion sections, held office hours, and graded assignments.	Fall 2022
Teaching Assistant <i>CSE11: Accelerated Intro to Programming</i> - Took responsibility for one week of the course in the instructor's absence, leading 2 lectures, and giving an online midterm exam to over 200 students. - Led the weekly discussion sections, including writing review notes and practice problems tailored to the course, leading live discussions, and creating pre-recorded videos.	Spring 2020
Teaching Assistant <i>CSE131: Compiler Construction</i> - Assisted in the design of new course assignments, updated the compilation target of existing assignments from Intel x86 to x86_64, and created tests and autograding infrastructure.	Fall 2019

Service & Mentorship

Student Organizer	PLATEAU 2024
External Reviewer	CHI 2023
Program Committee Member	HATRA 2023
Student Volunteer	PLDI 2023
GradWIC Mentor	2022 - 2023
ERSP Mentor	2021 - 2022